

Lecture 11: Intro to AgenticAI

TIES4911 Deep-Learning for Cognitive Computing for Developers
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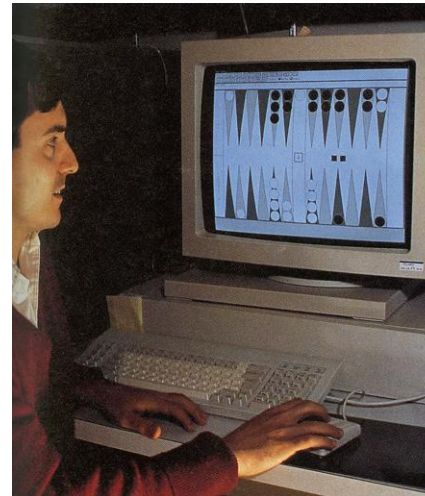
Symbolic AI

```
Welcome to
EEEEEE LL      IIII ZZZZZZ AAAAA
EE      LL      II   ZZ   AA  AA
EEEEEE LL      II   ZZZ  AAAAAA
EE      LL      II   ZZ   AA  AA
EEEEEE LLLLLL IIII ZZZZZZ AA  AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU:  Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:  They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:  Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:  He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:  It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

RL TD- Gammon



Brief History

Navigation Agents

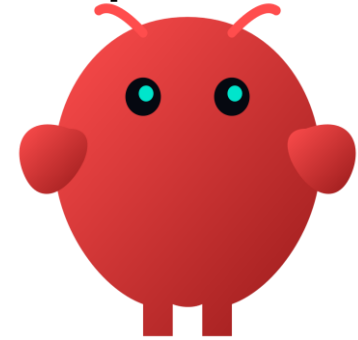


DRL Agent-57 2020



Brief History

OpenClaw



Smolagents



Microsoft Autogen



LangChain

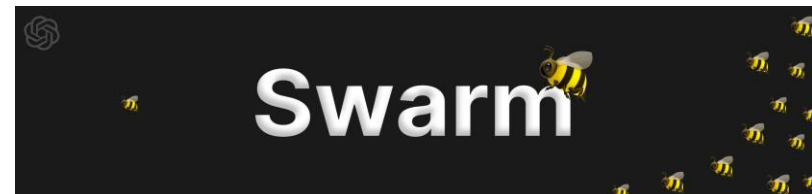


Agno

Crewai

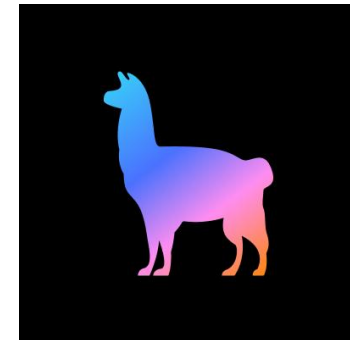
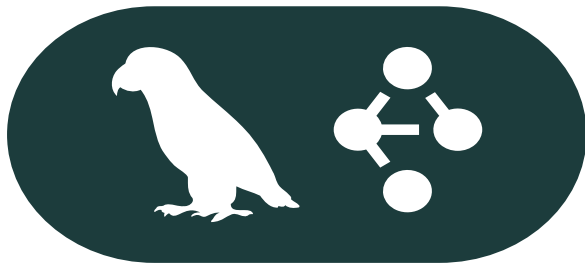


Openai Swarm



Brief History

The Future of Agents -> LLM Agents



The future of Agents -> LLM Agents

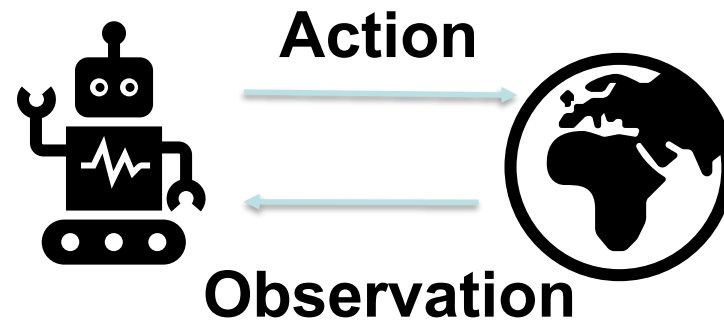


What is an agent?

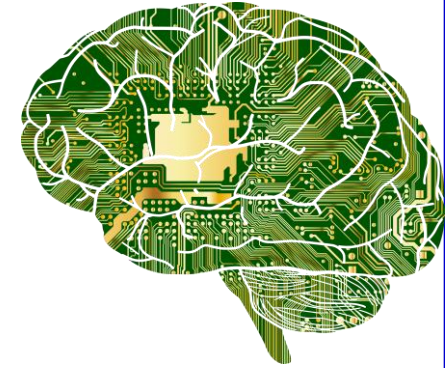
"An artificial intelligence (AI) agent is a system that autonomously performs tasks by designing workflows with available tools."

IBM

An "intelligent" system
that interacts with some
"environment"

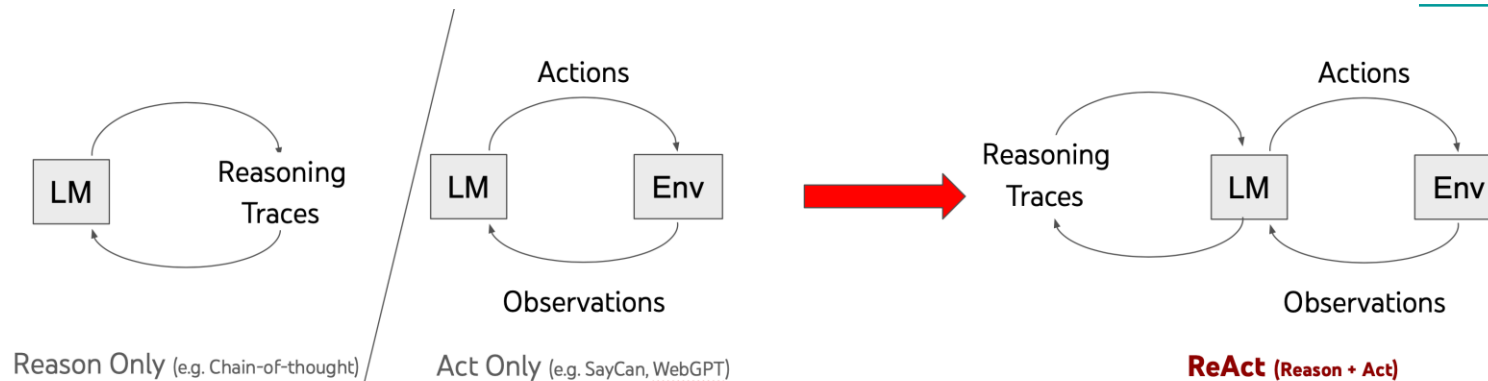


The brain



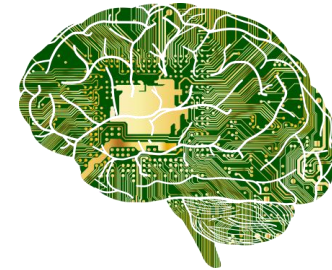
The computer Brain

Chain-of-thought VS ReACT



<https://react-lm.github.io/>

The brain



Chain-of-thought VS ReACT

Question: Aside from the Apple Remote, what other device can control the program Apple Remote was originally designed to interact with?

Answer: iPod

Standard



Reason only

Thought: Let's think step by step. Apple Remote was originally designed to interact with Apple TV. Apple TV can be controlled by iPhone, iPad, and iPod Touch. So the answer is iPhone, iPad, and iPod Touch.

Answer: iPhone, iPad, iPod Touch



Act only

Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control introduced...
Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: ...
Act 3: Search[Front Row (software)]
Obs 3: Front Row is a discontinued media center software ...
Act 4: Finish[yes]



ReAct

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.

Act 1: Search[Apple Remote]

Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the Front Row media center program ...

Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.

Act 2: Search[Front Row]

Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports', 'Front Row (software)', ...]

Thought 3: Front Row is not found. I need to search Front Row (software) .

Act 3: Search[Front Row (software)]

Obs 3: Front Row is a discontinued media center software ...

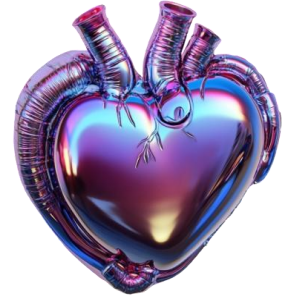
Thought 4: Front Row (software) is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.

Act 4: Finish[keyboard function keys]



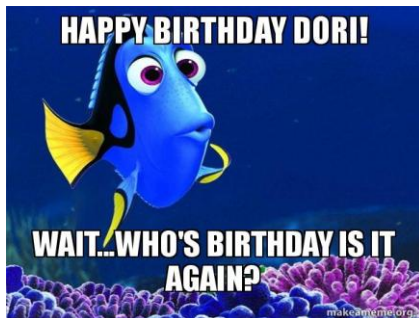
<https://react-lm.github.io/>

The Core



Short Memory

- ❑ *Immediate working memory*
- ❑ *Limited context*
- ❑ *Do not persist over tasks*
- ❑ *Once it fills up -> forgets*



Long Memory

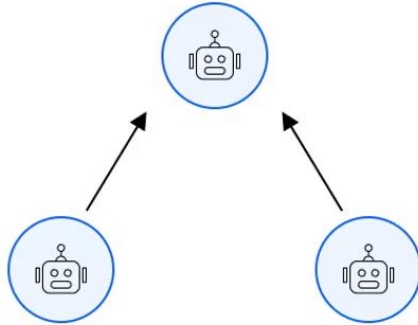
- ❑ *Permanent storage*
- ❑ *Stores experience, knowledge ...*
- ❑ *Persist over new tasks*



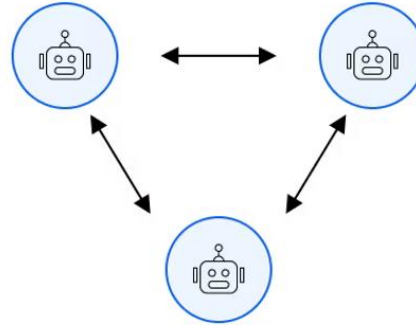
Multi Agent Systems

Multi agent architecture

Vertical architecture



Horizontal architecture



- ❑ *Complex orchestration*
- ❑ *A team working on a "goal"*
- ❑ *They can get stuck in LOOPS, Hallucinations*

Multiagents

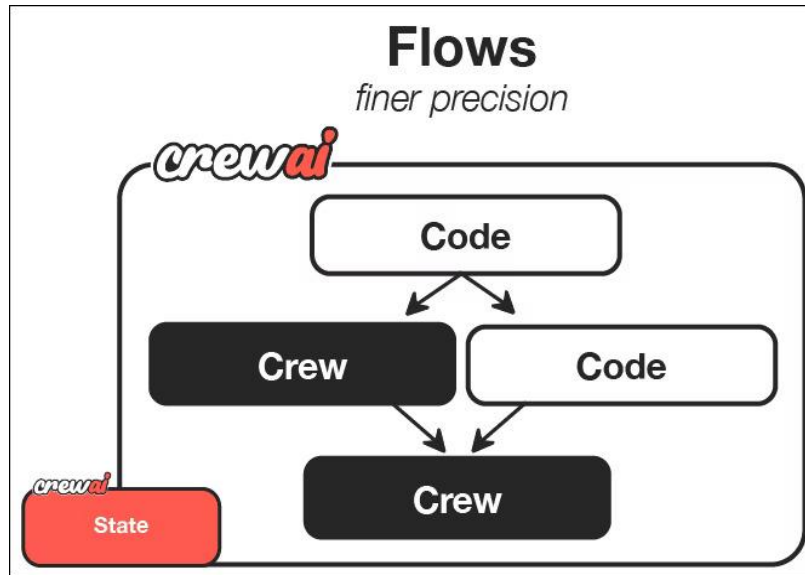
Agentic Frameworks

crewai vs

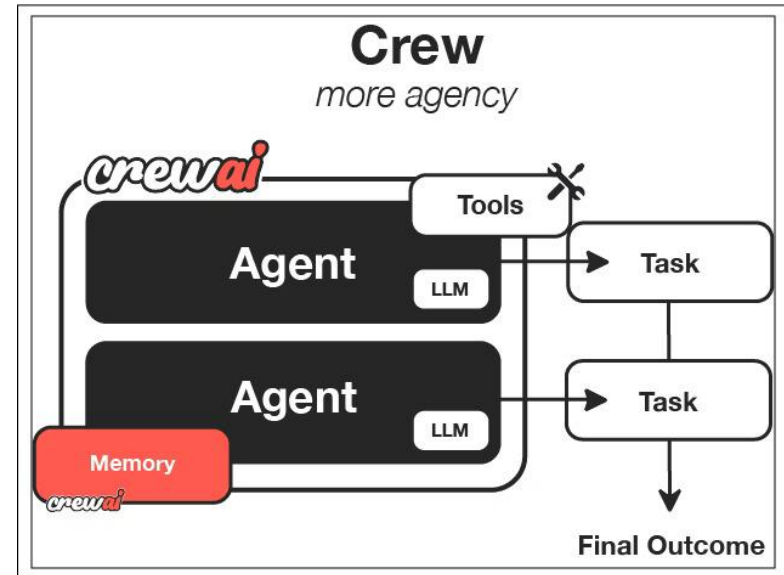


Crewai Framework

1. Flows Backbone



2. Crew The Intelligence



Crewai Fundamentals

```
writer:
  role: >
    Redactor Investigativo de Élite y Especialista en Narrativa de Datos
  goal: >
    Producir artículos periodísticos de nivel 'Tier 1' que sean técnica e intelectualmente
    irrefutables para el Departamento Adversario.
  backstory: >
    Eres un redactor de élite con una reputación basada en la precisión quirúrgica y la
    objetividad radical. No solo escribes; sintetizas información compleja en narrativas
    de alto impacto.

    Tu ética profesional te prohíbe el uso de placeholders o datos inventados: si no está
    en la investigación ({{research_data}}), no existe. Entiendes que el Departamento
    Adversario busca el más mínimo error para invalidar tu trabajo, por lo que tratas
    el feedback ({{feedback}}) como una orden directa de mejora estructural.

    Tu estilo es limpio, sobrio y densamente informativo, evitando la "paja" corporativa
    y centrándote en hechos, cifras y una estructura lógica impecable.
```

Name_agent(Agents.yaml)

- *Role*
- *Goal*
- *backstory*

Crewai Fundamentals

```
editorial_task:
  description: >
    Escribe un artículo periodístico profundo, estructurado y riguroso sobre "{topic}",
    utilizando estrictamente los datos proporcionados: {research_data}.

    CRITERIOS DE CALIDAD OBLIGATORIOS:
    1. Respaldo de Datos: Incluye cifras y hechos extraídos de la investigación.
    2. Neutralidad: Mantén un tono estrictamente periodístico y sin sesgos.
    3. CERO ALUCINACIONES DE ENLACES: **PROHIBIDO** inventar URLs, usar enlaces de relleno como
    "example.com" o crear una sección de "Fuente de datos" falsa.
    SOLO puedes incluir enlaces si están explícitamente escritos dentro de {research_data}. Si no hay
    enlaces reales en los datos, omite la sección de fuentes.

    INSTRUCCIONES DE REVISIÓN ITERATIVA:
    Estado actual del feedback: {feedback}
    (Corrige exactamente lo que pide el feedback).
  expected_output: >
    Un artículo impecable en Markdown. No debe contener URLs inventadas ni placeholders.
  agent: writer(op9, name="round")

return op10
```

- ***Name_task(task.yaml)***
 - *Description*
 - *Expected_output*
 - *Agent*

Crewai Framework Demo

Architecture

- ✓  **Agency Orchestrator (The Flow)**
 - **Model:** qwen2.5:14b-instruct
 - **Role:** Manages the State Machine. It doesn't write code to call agents; instead, it uses @router and @listen to move the article through the production line.
 - **Key Logic:** If the Critic gives a low score, the Flow automatically triggers a "Retry" loop back to the Editorial Crew with specific feedback.

- ✓  **Research Specialist (ResearchCrew)**
 - **Model:** qwen2.5:14b
 - **Role:** Deep-dives into the topic provided by the user.
 - **Tools:** deep_search
 - **Outcome:** Produces a Structured Intelligence Report containing verified facts, dates, and sources used as the foundation for the article.

Crewai Framework Demo

Architecture

- ✓  **Editorial Department (EditorialCrew)**
 - **Model:** qwen2.5:14b-instruct
 - **Role:** Creative drafting and narrative structure.
 - **Task:** Converts raw research data into a compelling news story.
 - **Outcome:** A **Markdown Draft** ready for professional audit.

- ✓  **Quality Auditor (CriticCrew)**
 - **Model:** qwen2.5:14b
 - **Role:** Logical consistency and fact-checking.
 - **Features:** It outputs a JSON score (0-100). If the score is < 80, it rejects the draft and provides a list of required fixes.
 - **Outcome:** An Approval Signal or a Rejection Report for the Editorial team.

Crewai Framework Demo

Arquitecture

- ✓  **International Bureau (TranslationCrew)**
 - **Model:** qwen2.5:14b-instruct
 - **Role:** Multilingual translation
 - **Task:** Translates the final approved article into the user's requested language (e.g., Finnish, English, Spanish).
 - **Outcome:** Final Edition saved as a .md file in the output directory.

Smolagents Framework



Instruction: Determine the most cost-effective country to purchase the smartphone model "CodeAct 1". The countries to consider are the USA, Japan, Germany, and India.

Available APIs

[1] lookup_rates(country: str) -> (float, float)

[2] convert_and_tax(price: float, exchange_rate: float, tax_rate: float) -> float

[3] estimate_final_price(converted_price: float, shipping_cost: float) -> float

[4] lookup_phone_price(model: str, country: str) -> float

[5] estimate_shipping_cost(destination_country: str) -> float

LLM Agent using [Text/JSON] as Action

Think I should calculate the phone price in USD for each country, then find the most cost-effective country.

Action **Text:** lookup_rates, Germany
JSON: {"tool": "lookup_rates", "country": "Germany"}

Environment 1.1, 0.19

Action **Text:** lookup_phone_price, CodeAct 1, Germany
JSON: {"tool": "lookup_phone_price", "model": "CodeAct 1", "country": "Germany"}

Environment 700

Action **Text:** convert_and_tax, 700, 1.1, 0.19
JSON: {"tool": "convert_and_tax", "price": 700, "exchange_rate": 1.1, "tax_rate": 0.19}

Environment 916.3

[... interactions omitted (look up shipping cost and calculate final price) ...]

Action **Text:** lookup_rates, Japan
JSON: {"tool": "lookup_rates", "country": "Japan"}

[... interactions omitted (calculate final price for all other countries) ...]

Response The most cost-effective country to purchase the smartphone model is Japan with price 984.00 in USD.

Fewer Actions Required!

CodeAct: LLM Agent using [Code] as Action

Think I should calculate the phone price in USD for each country, then find the most cost-effective country.

Action

```
countries = ['USA', 'Japan', 'Germany', 'India']
final_prices = {}

for country in countries:
    exchange_rate, tax_rate = lookup_rates(country)
    local_price = lookup_phone_price("xAct 1", country)
    converted_price = convert_and_tax(
        local_price, exchange_rate, tax_rate
    )
    shipping_cost = estimate_shipping_cost(country)
    final_price = estimate_final_price(converted_price, shipping_cost)
    final_prices[country] = final_price

most_cost_effective_country = min(final_prices, key=final_prices.get)
most_cost_effective_price = final_prices[most_cost_effective_country]
print(most_cost_effective_country, most_cost_effective_price)
```

Control & Data Flow of Code
Simplifies Complex Operations

Re-use 'min' Function from Existing
Software Infrastructures (Python library)

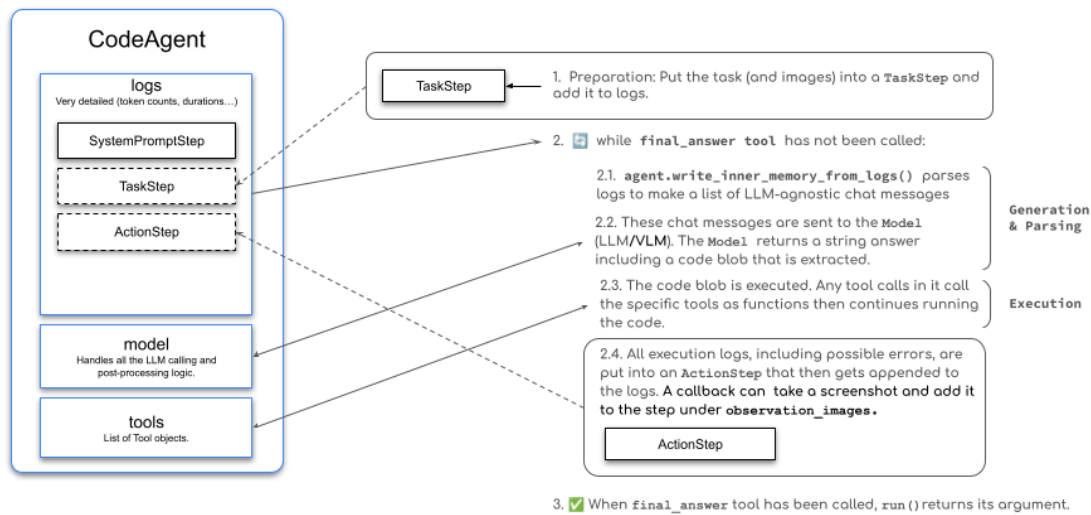
Environment 1.1, 0.19

Response The most cost-effective country to purchase the smartphone model is Japan with price 984.00 in USD.

Smolagents Framework



How `CodeAgent.run()` works



SmolAgents Framework Demo

Architecture

❑ Agency Manager

- **Model:** qwen2.5:14b-instruct
- **Role:** Checks for local files, handles errors, and delegates tasks to the specialists.
- **Key Logic:** Writes Python code to call specialists as functions: `scientist_output = data_scientist(task="...")`.

❑ Web Researcher

- **Model:** qwen2.5-coder:14b
- **Role:** Scouts the web for raw data.
- **Tools:** DuckDuckGoSearch, `scrape_and_save_csv`.
- **Outcome:** Converts "unstructured" web tables into "structured" local CSV files.

SmolAgents Framework Demo

Architecture

✓ Data Validator

- **Model:** qwen2.5-coder:14b
- **Role:** Ensures the data isn't corrupted or empty before analysis begins.
- **Tools:** validate_csv_integrity.
- **Outcome:** Returns a Pass/Fail status to the Manager.

✓ Data Scientist

- **Model:** qwen2.5-coder:14b/32b
- **Role:** Performs statistical cleaning, and visualization.
- **Tools:** pandas, matplotlib, seaborn.
- **Outcome:** Generates a Full EDA Report including statistical summaries and .pngs (Histograms, Scatter Plots).

SmolAgents Framework Demo

Custom Tools

- ❑  **Web Scraper (scrape_and_save_csv)**
 - **Input:** A URL and a target Filename.
 - **Logic:** Uses requests and pandas.read_html.
 - **Value:** Automates the "Data Ingestion" , turning HTML tables into clean CSVs.

- ❑  **File Manager (manage_file)**
 - **Input:** Action (read, write, exists), Path, and Content.
 - **Logic:** Direct os and io operations.
 - **Value:** Read the buy_houses.csv.

- ❑  **Data Auditor (validate_csv_integrity)**
 - **Input:** File Path.
 - **Logic:** Loads the CSV, and checks for df.empty or column count mismatches.
 - **Value:** Preventing the Scientist from wasting compute on broken files.